AI for Healthcare

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Project Report

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1. Introduction

Artificial Intelligence has become one of the most influential technologies in modern healthcare. With the increasing availability of medical imaging data and advancements in deep learning algorithms, AI systems can assist healthcare professionals in detecting diseases more accurately and efficiently. Among various medical applications, skin disease diagnosis is particularly well suited for computer vision because dermatological conditions can often be identified through visual examination of skin lesions.

Skin diseases affect millions of individuals worldwide and range from harmless benign lesions to aggressive malignant cancers such as melanoma. Early detection significantly improves treatment success and patient survival. However, accurate diagnosis requires trained dermatologists, whose availability may be limited, especially in rural and underserved regions.

Deep learning models have demonstrated remarkable performance in medical image classification. Convolutional Neural Networks (CNNs) automatically learn complex visual features directly from images, eliminating the need for handcrafted feature extraction techniques. Modern architectures such as EfficientNet provide high classification accuracy while maintaining computational efficiency, making them suitable for real-world healthcare applications.

This project develops an AI-powered healthcare assistant capable of classifying dermoscopic skin lesion images into eight clinically relevant disease categories. The system not only predicts the disease but also enhances interpretability by highlighting important image regions using Grad-CAM. Furthermore, an AI language model generates simple, understandable explanations of the predictions, making the application more accessible to non-technical users.

The project demonstrates the integration of deep learning, explainable AI, and natural language generation into a practical healthcare application through a web-based interface built using Streamlit.

2. Problem Statement

Skin diseases are among the most common medical conditions worldwide. While many skin lesions are benign, some conditions such as melanoma require immediate medical attention. Early diagnosis greatly increases the chances of successful treatment. Unfortunately, accurate diagnosis often depends on experienced dermatologists and specialized diagnostic equipment, which are not universally available.

Several challenges exist in traditional skin disease diagnosis:

- Limited availability of dermatologists in many regions.
- Subjective interpretation of skin lesions.
- High patient-to-doctor ratio leading to delayed diagnosis.
- Lack of accessible preliminary screening tools for the general public.

The objective of this project is to design an AI-powered healthcare assistant capable of analyzing dermoscopic skin images and predicting the most probable disease category using deep learning techniques. The application should also provide an explainable visualization of the model's decision-making process and generate a human-readable explanation of the prediction.

3. Objectives

The primary objectives of this project are:

- Develop a deep learning model capable of classifying skin lesion images into eight disease categories.
- Train the model using the ISIC 2019 Skin Lesion Dataset.
- Achieve reliable classification accuracy while maintaining computational efficiency.
- Improve model interpretability using Grad-CAM visualizations.
- Generate user-friendly AI explanations using a Large Language Model through OpenRouter.
- Develop an interactive Streamlit web application for real-time inference.
- Demonstrate the practical application of deep learning in AI-assisted healthcare.

4. Dataset Description

The model was trained using the **ISIC 2019 Skin Lesion Dataset**, one of the most widely used benchmark datasets for skin disease classification.

Dataset Statistics

Property	Value

Dataset	ISIC 2019
Total Images	25,331
Image Type	Dermoscopic Images
Classes	8

Disease Categories

Class	Disease
AK	Actinic Keratosis
BCC	Basal Cell Carcinoma
BKL	Benign Keratosis
DF	Dermatofibroma
MEL	Melanoma
NV	Melanocytic Nevus
SCC	Squamous Cell Carcinoma
VASC	Vascular Lesion

The images were organized into separate folders based on disease category, making them suitable for supervised learning.

Data Distribution

The dataset contains an imbalanced class distribution. The Nevus (NV) category contains the highest number of images, whereas Dermatofibroma (DF) and Vascular lesions (VASC) contain relatively fewer samples. This imbalance presents additional challenges during model training and evaluation.

5. Methodology

The complete workflow of the project consists of multiple stages.

Step 1: Data Collection

The ISIC 2019 dataset was downloaded and organized into class-wise directories.

Step 2: Data Preprocessing

Before training, all images were resized to **224 × 224 pixels** to match the input size required by EfficientNet-B0.

The preprocessing pipeline included:

- Image resizing
- Tensor conversion
- Pixel normalization
- Data augmentation (random horizontal flip and random rotation)

Step 3: Dataset Splitting

The dataset was divided into:

- Training Set
- Validation Set
- Test Set

This ensures unbiased evaluation of model performance.

Step 4: Model Training

EfficientNet-B0 pretrained on ImageNet was fine-tuned using transfer learning.

The final classification layer was modified to predict eight disease categories.

Training was performed using:

- Optimizer: Adam
- Loss Function: CrossEntropyLoss
- Epochs: 8
- Learning Rate: 0.001

Step 5: Explainability

Grad-CAM was integrated to highlight the image regions responsible for model predictions.

Step 6: AI Explanation

The predicted disease, confidence score, severity level, and recommendation were sent to an OpenRouter LLM, which generated a simple explanation for users.

8. Experimental Results

The trained EfficientNet-B0 model achieved the following performance after eight training epochs.

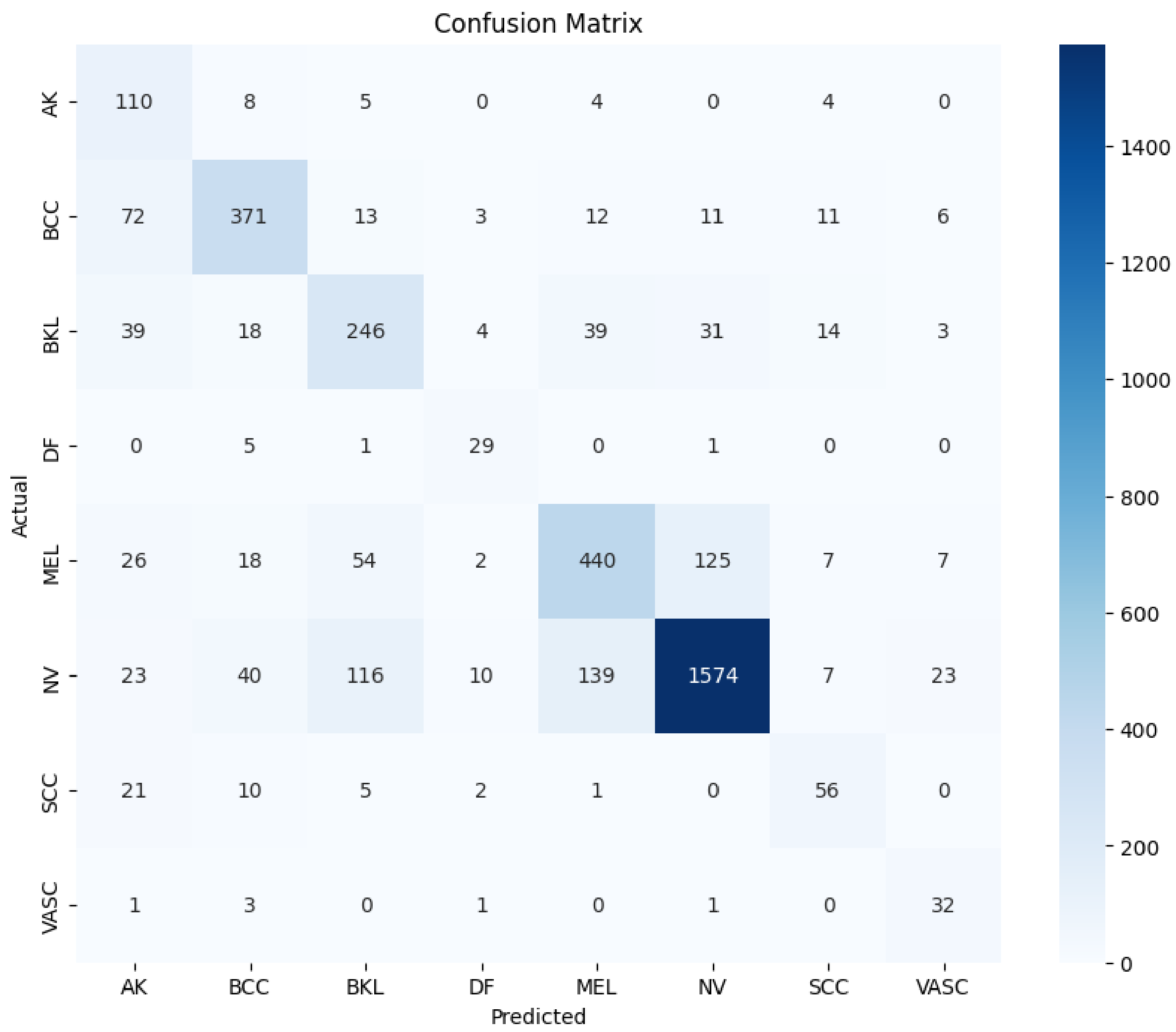
Training Performance

Metric	Value
Training Accuracy	76.66%
Validation Accuracy	74.78%
Test Accuracy	75.13%

The classification report obtained on the test dataset is summarized below.

Disease	Precision	Recall	F1-score
AK	0.38	0.84	0.52
BCC	0.78	0.74	0.76
BKL	0.56	0.62	0.59
DF	0.57	0.81	0.67
MEL	0.69	0.65	0.67
NV	0.90	0.81	0.86
SCC	0.57	0.59	0.58
VASC	0.45	0.84	0.59

The model demonstrated strong performance on common disease categories such as Melanocytic Nevus while maintaining acceptable performance across other disease classes despite dataset imbalance.

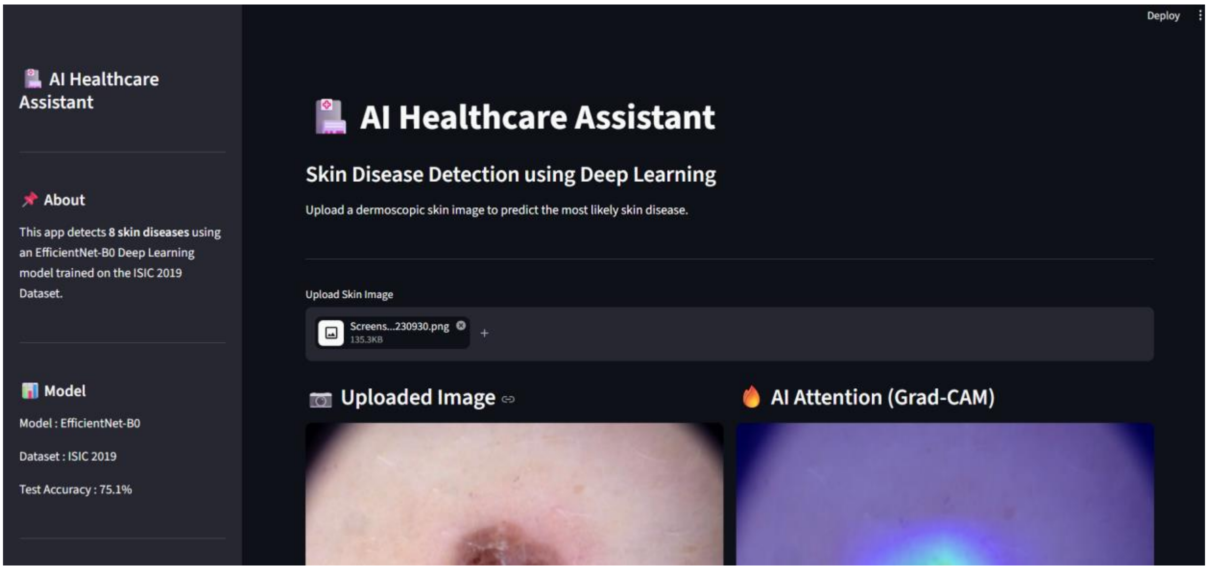


6. Streamlit Application

A user-friendly web application was developed using Streamlit.

The application provides the following features:

- Upload skin lesion images.
- Perform real-time disease prediction.
- Display prediction confidence.
- Show the top three predicted disease classes.
- Visualize model attention using Grad-CAM.
- Generate AI-powered explanations using OpenRouter.
- Display medical recommendations and precautions.



The Streamlit interface makes the system accessible to users without requiring technical knowledge.